



A Review of the Pore Pressure Gradients at Pharos-1

WA-398-P

Client: ConocoPhillips Australia

Report Prepared By: JON ELLIOTT

AIPC Pore Pressure Engineering

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1 Introduction

Note: All formation tops stated in the following report are PRELIMINARY picks made in the field. For final stratigraphy please refer to ConocoPhillips Well Completion Report Volume 2: Interpretive Data; Section 2.4.

AIPC Pty Ltd conducted a review of the distribution of pore pressure and fracture gradients in the vicinity of WA-398-P prior to the drilling of Pharos-1. The study involved the modelling of shale porosity from deep resistivity and sonic logs from the offset wells to locate the stratigraphic distribution and magnitude of the pore pressure gradients in the vicinity of the Pharos prospect. The objective of the study was to establish compaction trends and Eaton coefficients that might be applied during the drilling of Pharos-1 and other wells in the area. Fracture gradient modelling in these wells was also conducted and calibrated with the available leak-off test data. Wells reviewed in the study include;

- | | |
|-------------------|--------------|
| • Caswell 2 | • Kontiki-1 |
| • Buffon-1 | • Poseidon-2 |
| • Poseidon-1 | • Kronos-1 |
| • Gorgonichthys-1 | • Kronos-1 |
| • Titanichthys-1 | • Zepyros-1 |
| • Abalone-1 | • Boreas-1 |
| • Turbo-1 | • Grace-1 |
| • Scott Reef-1 | |

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During the drilling of Pharos-1 the available LWD logs and corrected drilling exponent (Dxc) were modelled to quantitatively determine the pore pressure gradients encountered in “real-time”. Pore pressure monitoring was conducted remotely on a daily and “as-needs” basis during critical stages of the well. LWD deep resistivity in the 12 ¼”, 8 ½” and 6 ½” hole sections while sonic was also acquired in the 8 ½” and 6 ½” hole sections. The gamma ray response, in conjunction with cuttings description and the mudlog were used to determine shale content in the section.

The results of the pre-well modelling using the results from the wells listed above suggested that there may not be a normally pressured shale section from which to extrapolate normal compaction trend lines to the Jamieson Formation and the section below the Jamieson Formation. Above the Jamieson Formation the section penetrated was predominated by carbonates while the sonic velocities at the top of the Cenomanian aged Jamieson Formation are considerably too slow to be considered normally pressured for the depth at which they were encountered. Work conducted by Phil Heppard, ConocoPhillips pore pressure specialist in Houston proposed trends to be applied for sonic and deep resistivity modelling purposes. The sonic trend proposed by Heppard is from Barossa-1, a COPA operated well located to in the Bonaparte Basin. Figure 1 below details compressional sonic transit times for the Jamieson Formation in the Browse Basin as well in other parts of the world in normally pressured shale predominant sections. The plot clearly illustrates the

rationale for proposing that the entire Jamieson Formation at Buffon-1 and wells in the Greater Poseidon area is overpressured from the uppermost part of the formation.

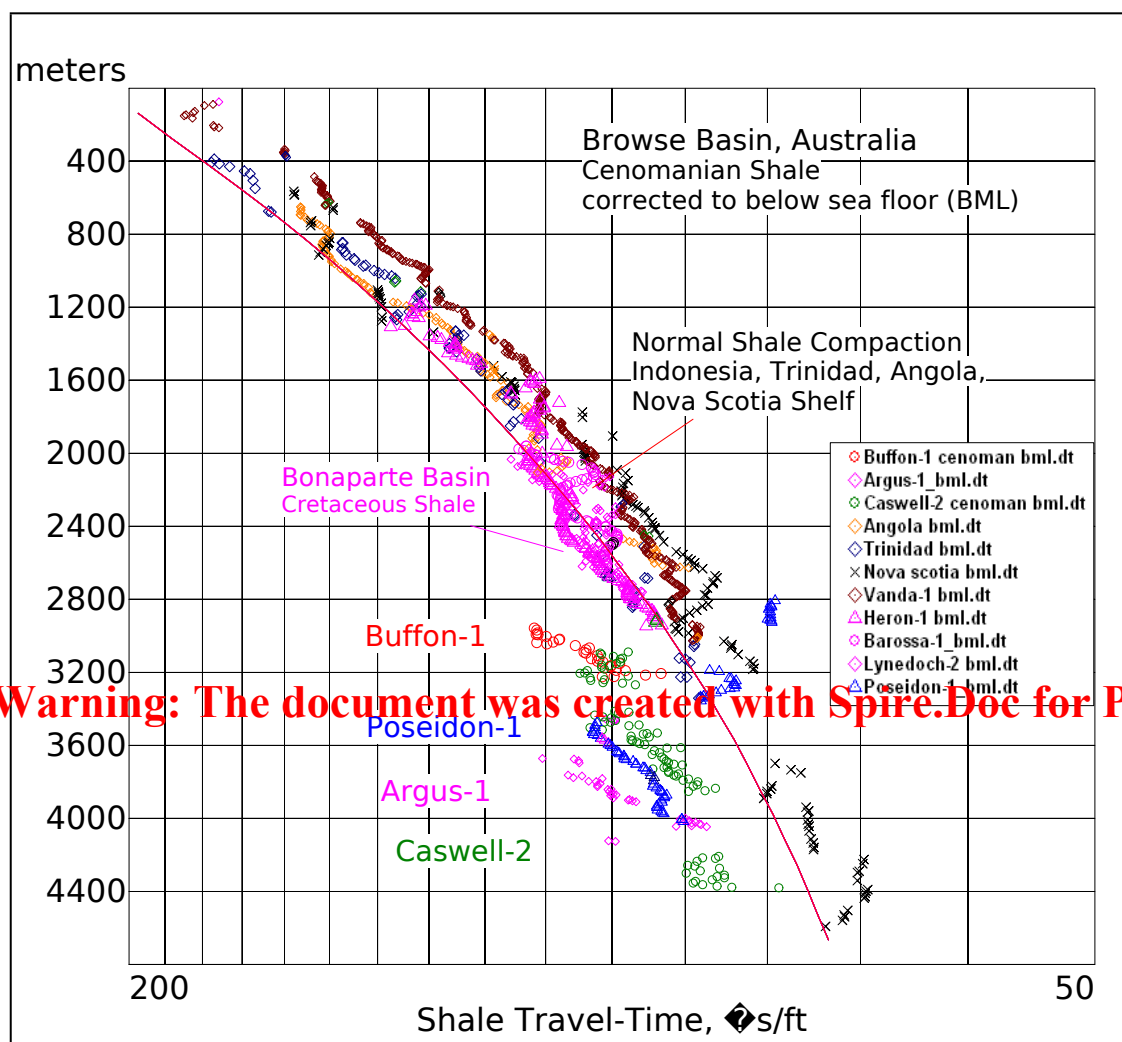


Figure 1: Proposed Compaction Trend for COPA wells in Browse Basin (Heppard)

The study indicates that the top of the Jamieson Formation is overpressured in most wells in the deeper sections of the Browse Basin and there are very few intervals of normally compacted shales in the area that might allow for the extrapolation of normal compaction trends. After an examination of a number of offset wells a small section of compressional sonic in the shales at the top of the Jamieson Formation at Gorgonichthys-1 (Inpex 2000) was identified as being normally pressured. The extrapolated sonic compaction trend from Gorgonichthys-1 was close to the proposed Bonaparte sonic trend both in magnitude and gradient. In light of this observation, the Gorgonichthys-1 sonic trend has been used in this study and finessed to match any calibration

points in Pharos-1. The resistivity compaction trend applied to Pharos-1 was modified from the deep resistivity compaction trend interpreted from Gorgonichthys-1.

In the case of Pharos-1, a normally compacted calcareous shale interval was encountered from 3710 to 3800m as defined from the deep resistivity response. This supports the compaction trend proposed by Heppard and the trend extrapolated from Gorgonichthys.

2 Methodology

It is generally thought that the majority of instances of overpressure in the world are caused by the undercompaction of shales. This is thought to be the situation in the overpressured section at Pharos-1 as well as in many of the offset wells in the Browse Basin including Buffon-1 (Woodside 1980) and wells drilled in the Greater Poseidon area by ConocoPhillips and its partners since 2008. When shale-predominant intervals cannot progressively expel the trapped water as the depth of burial and the weight of overburden increases, the pore fluid starts to take on a portion of the overburden weight. When the pore pressure exceeds the pressure that might be exerted by a continuous column of water from the surface, the formation is said to be overpressured.

Pore pressure and fracture gradient (PPFG) modelling at Pharos-1 was conducted using the measured log responses and calibrated using mudlog and engineering indications.

2.1 PPFG Determination from Wireline and LWD Measurements

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 A normally compacted sequence of shales has progressively reduction in porosity with depth. The understanding of the porosity of the intervals of shale is therefore critical in understanding undercompaction overpressure. The two electric log responses that are traditionally used for shale porosity determination are compressional sonic and deep resistivity.

The compressional sonic transit times tend to become progressively faster with depth through a sequence of shales as the porosity decreases. If there is departure to slower transit times with depth in a massive shale interval this is an indication of an increasing pore pressure gradient. In the same way, the deep resistivity increases with progressive compaction of shales. Departure to lower resistivity in a shale sequence is an indication of an increasing volume of conductive water in the shale and also of an increasing pore pressure gradient.

Eaton (1975) proposed that an estimate of pore pressure was possible if the rate of porosity decay in normally pressured shales was known (i.e. by defining a normal compaction trend line). Other inputs to the Eaton equation are the overburden gradient, the hydrostatic gradient and a coefficient “n”, which is the Eaton coefficient. The Eaton equation is as follows;

$$P = OBG - (OBG - NHG) * (R_o / R_n)^n$$

Where **P** is the pore pressure, **R_o** is observed resistivity, **R_n** is normal resistivity, **OBG** is the overburden gradient, **NHG** is the normal hydrostatic gradient, **R_o** is the observed resistivity and **n** is an empirically derived exponent. This equation is also applied for use with the sonic log, Dxc, seismic velocity and conductivity.

For this study, fracture gradients have been derived using the Eaton Equation with Poisson's ratios estimated from well data, specifically leak-off test data from the 14 wells that have been reviewed in the study. The Eaton fracture gradient estimation equation is as follows;

$$FG = PP + (OBG - PP) * (\nu / (1 - \nu))$$

Where **FG** is the fracture gradient, **PP** is the formation fluid pressure at the point of interest, **OBG** is the overburden gradient, and ν is Poisson's ratio.

At Pharos-1 the overburden gradient was calculated using the available LWD density data while the density curve from Buffon-1 was used for intervals where no density data was available. Shale determination was conducted using the gamma ray log in conjunction with the mudlog.

2.2 Mudlog Pore Pressure Evaluation

Mudlog indications of overpressure are still of paramount importance in the evaluation of the magnitude of overpressure. When a well is being drilled underbalanced, gas tends to percolate into the borehole whilst making each connection. Gas peaks recorded a lag time after a connection are referred to as connection gases and the first occurrence of a connection gas is a good calibration point in the construction of a pore pressure model as it indicates that a part of the well is being drilled underbalanced at that depth. The identification of the first connection gas is of extra significance as it can offer a calibration point where the pore pressure first exceeds the mud hydrostatic.

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Other mudlog indications that the well is being drilled underbalanced are the occurrence of overpressured shale cavings, a general increase in background gas and an increase in the rate of penetration through a uniform sequence of shales. When shale sequences are drilled underbalanced, shale cavings tend to show a dramatic increase in volume and size and may be splintery, propeller or convex shaped as trapped pore fluids and gas within the shale forcibly expel them into the wellbore due to the fact that the formation pore pressure exceeds the wellbore hydrostatic pressure. The recognition of overpressured shale cavings is an integral part in the identification of overpressure and is generally an excellent indication that a section of the well is being drilled underbalanced.

Increasing background gas through a uniform sequence of shales may also be an indication of rising pore pressure particularly where it is unrelated to changes in the rate of penetration or lithology.

Other engineering issues such as kicks and lost circulation can be useful in the calibration of any pore pressure model.

3 Pharos-1 Pore Pressure Summary

Pore pressure monitoring was conducted remotely while drilling by AIPC from the uppermost part of the 12 ¼" hole section to TD in the 8 ½" hole section. The objectives of the pore pressure and fracture gradient modelling at Pharos-1 were to estimate the pore pressure and fracture gradients through the overpressured section and, if possible, to identify a pressure reversal known to be present above the Plover Formation. The identification of the pressure reversal was seen as a critical aspect of the well design as setting the 7-5/8" casing at an optimal depth would minimise

the risk of catastrophic mud losses due to excessive overbalance in a normally pressured reservoir section while excluding an interval of overpressured shale in the 6 ½" hole section. The 6 ½" section was evaluated post well and the findings from this part of the well is incorporated in this report. Note that all depths detailing in this summary are measured depth unless stated otherwise and that the well was essentially vertical.

Table 1 on the following page details the key formation tops encountered at Pharos-1.

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Formation Name	MDRT (m)	TVDRT (m)	TVDSS (m)	Thickness (m)
Baudin Marl Member	3077.5	3077.1	-3055.0	138.5
Johnson Formation	3216.0	3215.6	-3193.5	490.2
Wollaston Formation	3706.2	3706.2	-3683.6	303.8
Jamieson	4010.0	4009.4	-3987.3	466.0
Base Aptian/ Echuca Shoals Formation	4476.0	4475.4	-4453.3	309.0
Brewster	4785.0	4784.2	-4762.1	185.6
Montara	4970.6	4969.8	-4947.7	75.9
Plover A SST	5046.5	5045.7	-5023.6	110.5
Plover B SST	5157.0	5156.1	-5134.0	25.0
Triassic Sahul	5182.0	5181.1	-5159.0	38.0
TD	5220.0	5219.1	5181.1	-

Table 1: Pharos-1 Formation Tops From Baudin Marl to TD

3.1 Recent to Wollaston Formation (Seafloor-3706.2m)

Pharos-1 penetrated a top-hole section which comprised predominantly carbonates of recent to Early Palaeocene age. The interval from the seafloor to the top of the Wollaston Formation is thought to be entirely normally pressured and was drilled with seawater and gel sweeps. Losses were encountered throughout the 12-1/4" hole section to the 9-5/8" casing setting depth at 4035m. Low mudweights and low background gas levels, drilling fluid losses and an absence of any fluid influxes support the interpretation at the entire section in the well above the top of the Wollaston Formation is normally pressured. The nature of the carbonate predominant section suggests that the losses were to vugs and fractures within the carbonates. An LOT conducted at 2333m yielded a leak off of 11.7ppg EMW which is significantly below the modelled shale fracture gradient and is consistent with a brittle formation around the 13-3/8" casing shoe. The 13-3/8" casing was set in a carbonate predominant interval.

3.2 Wollaston Formation (3706.2-4010m)

The Wollaston Formation was encountered from 3706.2 to 4010m MD and comprised an interval of claystone and calcareous claystone overlying an interval of carbonate. The interval from 3706.2 to 3930m has a high gamma ray and has the petrophysical character of an interval of shale. The resistivity response from 3720-3800m lies close to the proposed normal compaction trend and this section is interpreted to be normally pressured. Low gas values averaging 0.07% are consistent with this interpretation. Losses experienced while drilling this section are most likely into intervals of carbonate higher in the open hole section. Below 3800m the resistivity shows a sharp drop and while this appears to also correspond to an increase in carbonate content, it is thought that 3800m represents the top of the transition zone. The pore pressure gradient at the base of the shale predominant interval within the Wollaston Formation at 3930m is estimated to be 12.5ppg EMW. From 3920m to the top of the Jamieson Formation at 4010m, the section is predominantly

carbonate and it is not possible to estimate pore pressure gradients through this interval. The pore pressure gradient in the uppermost Jamieson is estimated to be 13.7 ppg EMW.

3.3 Jamieson Formation (4010-4465m)

The Cenomanian aged Jamieson Formation was encountered from 4010 to 4465m. From 4010m to 4035m the Jamieson Formation was drilled in 12-1/4" hole section. Less than 10m of resistivity data was acquired at from in the 12 1/4" hole section from which a pore pressure gradient of 13.7 ppg EMW was estimated. The uppermost Jamieson Formation was drilled with seawater and gel sweeps however there were no indications of the 5 ppg EMW interpreted underbalance such as wellbore failure, elevated gas or connection gas peaks. The 9-5/8" casing was run at this point and cemented at 4029.5m. An LOT conducted after drilling to a measured depth of 4038m indicated a fracture gradient of 15.6 ppg EMW. This is in line with the modelled fracture gradient at this depth and pore pressure which is consistent with a Poisson's ratio of 0.375.

In the remainder of the Jamieson Formation above the Stokes Member the pore pressure gradients are interpreted to range from 12.2 to 13.7 ppg (1.47-1.65 SG) EMW from both sonic and deep resistivity responses through an interval of predominantly dark grey shale. There were no connection gases recorded while drilling the interval with a 13.5 ppg mud despite the interpreted underbalance in parts of the interval. This is consistent with other wells in the Greater Poseidon area and appears to be indicative of the absence of permeability in the underbalanced sections as well as the competency of the Jamieson Formation. In addition to an absence of connection gas peaks, there are no indications of wellbore failure as indicated on the caliper which was in gauge throughout the 8-1/2" hole section.

The Stokes Member, the lowermost unit of the Jamieson Formation at Pharos-1, was encountered from 4402-4465m and was represented by an interval of siltstone. The lithology encountered within the Stokes Member is not suitable for pore pressure gradient estimates however connection gases of 2.5% and 1.3% were recorded seemingly validating the estimates of pore pressure in the overlying Jamieson shales.

3.4 Echuca Shoals Formation (4465-4785m)

The Echuca Shoals Formation was encountered from 4465-4785m. The interval at Pharos-1 comprised massive medium to dark grey shale. The interval is interpreted to be overpressured with the deep resistivity response through the shales showing an increase with depth indicating a dropping pore pressure gradient. Immediately below the base of the Stokes Member however the departure from the NCTL is the greatest in the entire well defining the highest gradient at Pharos-1 of 14.0 ppg EMW at 4505m. This gradient is interpreted from both sonic and resistivity models and supported by the record of connection gases while drilling with a 13.5 ppg mud. The pore pressure gradient then drops with depth through the Echuca Shoals from 14.0 to 12.5 ppg EMW. The connection gas peaks show an increasing trend through the Echuca Shoals and the underlying Brewster Formation. It is not clear if there are further delivery points for the connection gases deeper in the 8 1/2" hole section other than the Stokes Member.

3.5 Brewster Formation (4785 to 4970.6m)

The interval from 4785 to 4950m was drilled in 8 ½" hole with the interpreted pore pressure gradient dropping from 12.5 to 11.0 ppg EMW. Connection gas peaks were recorded after each connection, the origin for which is assumed to have been predominantly the Stokes Member and the upper section of the Echuca Shoals Formation. A 7-5/8" liner was run at this depth and the well deepened in 6 ½" hole through to the base of the Brewster Formation at 4970.6m. No connection gases were encountered while drilling the Brewster Formation in the 6 ½" hole while drilling with a 9.5 ppg mud.

3.6 Montara Formation (4970.6 to 5046.5m)

A marked pressure reversal was penetrated within the upper part of the Montara Formation. A pre-test acquired in a lower Montara sand at 4987.9m had a pore pressure gradient of 8.58 ppg EMW indicating the pressure reversal is complete by this depth. There were no connection gases while drilling the Montara Formation. A leak-off test acquired at 4848.5m of 14.1 ppg EMW was marginally less than the estimated fracture gradient at this depth which may be an indication that the pore pressure gradient is less than the estimate from sonic and deep resistivity responses which was 10.7 ppg EMW. What does seem likely is that the 7-5/8" casing shoe was set within or immediately below the pressure reversal.

3.7 Plover Formation (5046.5 to 5182.0m)

The Plover Formation section encountered interbedded siltstones, sandstones and claystones. The section is not suitable for accurate pore pressure modelling from shale porosity estimates. Formation pressures acquired in the more permeable interval yielded pressure gradients that were normal and marginally above 8.5 ppg EMW. There are indications of a second transition zone commencing in the base of the Plover Formation with the reappearance of connection gases below 5175m.

3.8 Sahul Formation (5046.5 to 5182.0m)

The Triassic aged Sahul Formation was encountered from 5182-5220m. The section comprised interbeds of claystone, sandstone and siltstone and is not suitable for pore pressure estimation from shale porosity. A connection gas of 2.06% recorded at 5205m indicates that the pore pressure gradient exceeds 9.5 ppg EMW within this interval or possibly within the non-reservoir sections of the overlying Plover Formation.

The interpreted pore pressure profiles are detailed in Figure 2: Pharos-1 Deep Resistivity Equipoise Model and Figure 3: Pharos-1 Sonic Equipoise Model.